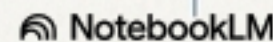


Software Architecture in the Era of Infinite Logic Generation



The 2026 Shift: From Generation to Curation

$$\text{Value} = \frac{\text{Reviewing Logic} \times \text{Architectural Integrity}}{\text{Writing Logic}}$$

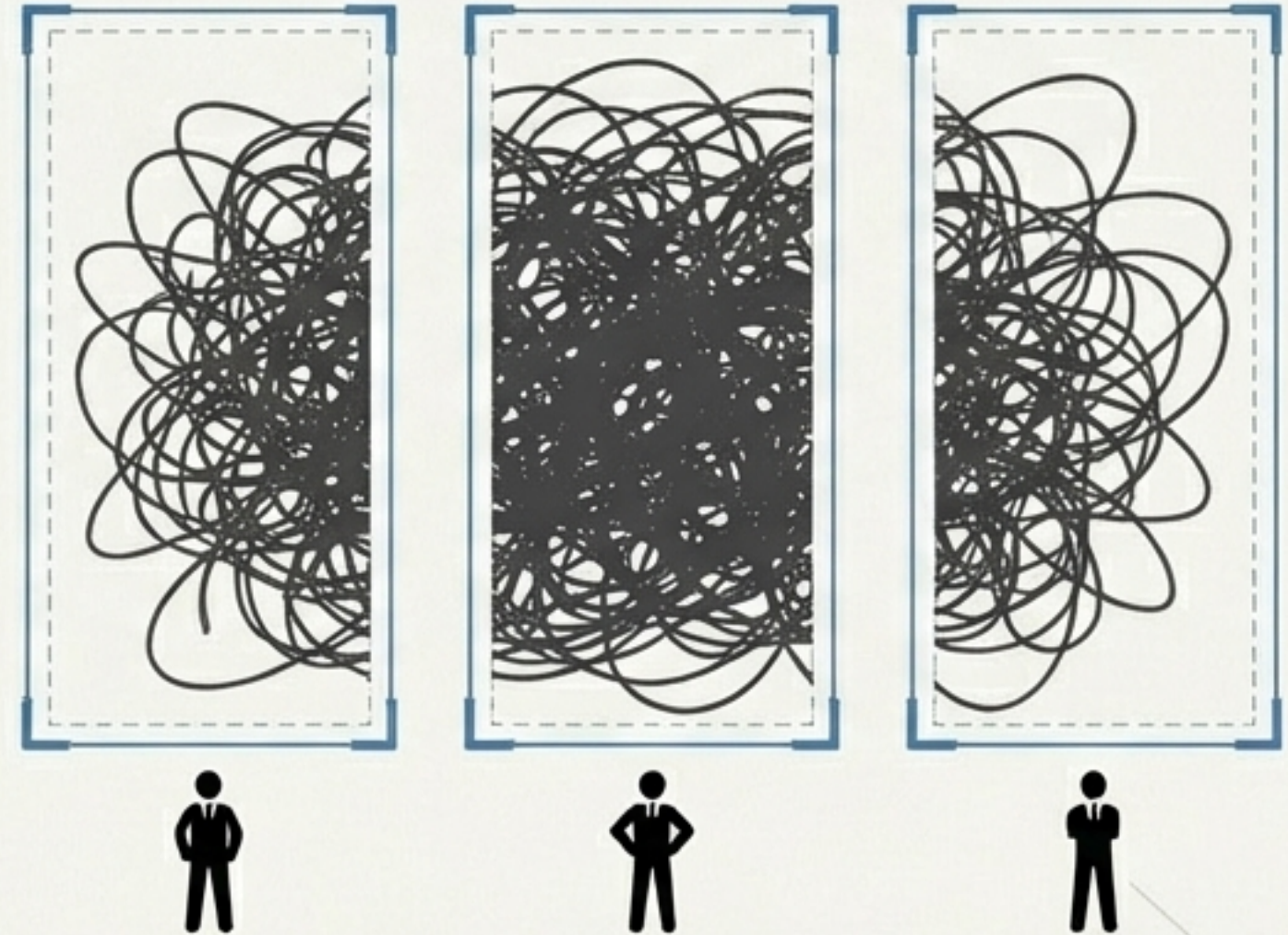

If an LLM can generate 10,000 lines of syntactically perfect code in seconds, the bottleneck is no longer typing. The bottleneck is structural comprehension.

The Original Sin of Work Division

Mind & Hands Limitation ⚠️



Divide and Conquer ✓



The problem provides the purpose to the expert knowledge. Expert knowledge is merely a resource to be managed based on problem needs.

The Restaurant Automation Case Study

- REQ-1: Support the work of wait & floor staff
JetBrains Mono
- REQ-2: Support the work of kitchen staff
JetBrains Mono
- REQ-3: Support the work of management
JetBrains Mono

Software-to-be



Host/Waiter

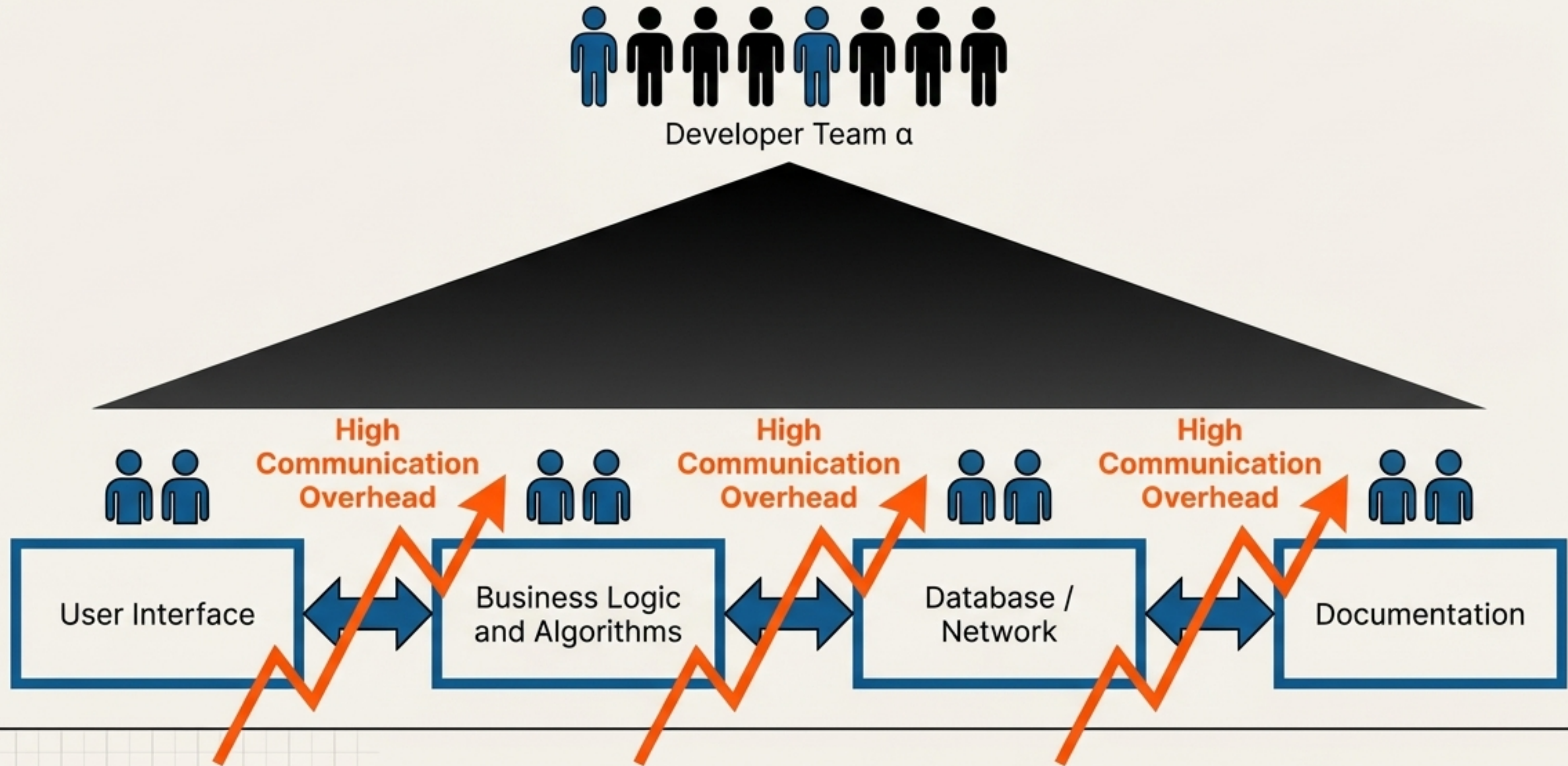


Chef



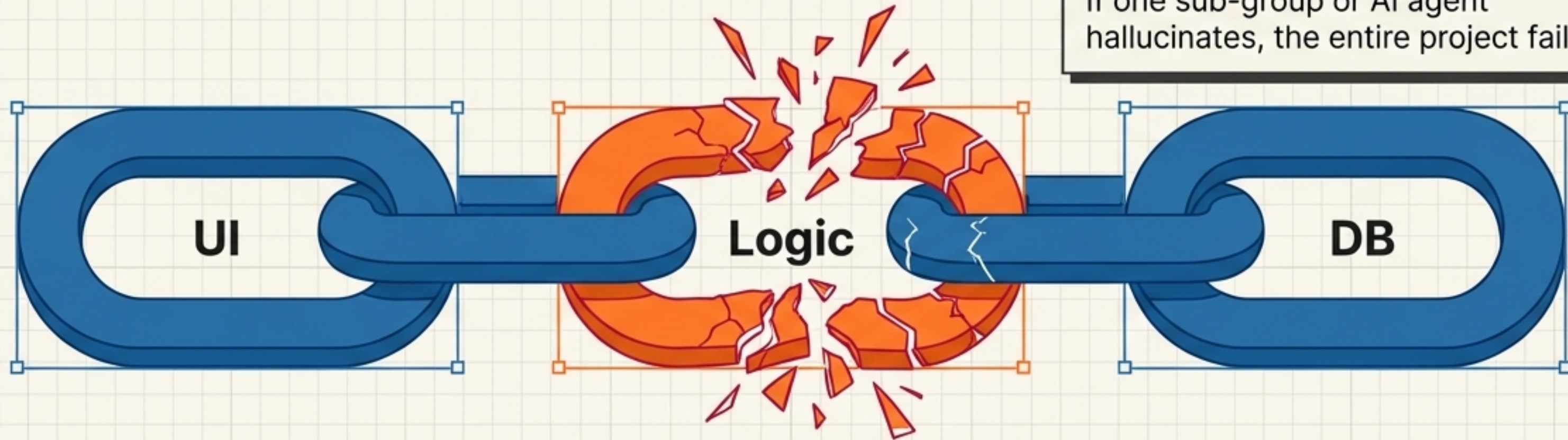
Manager

Team α : Contrived Horizontal Slicing



The Chain Vulnerability

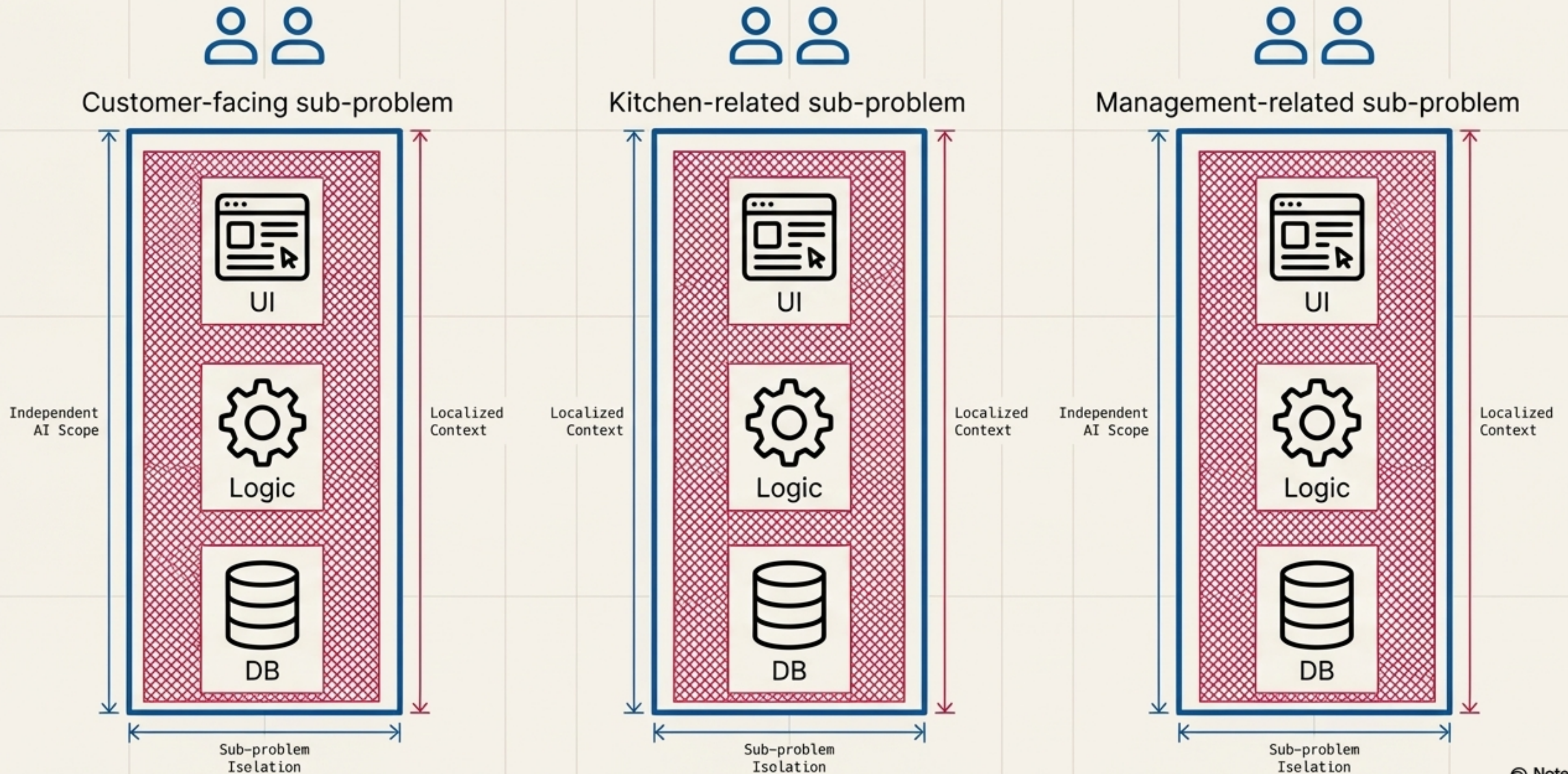
Failure Mode: Every link is critical. If one sub-group or AI agent hallucinates, the entire project fails.



```
> Prompt: "Write the business logic for the restaurant."
```

← "Contextual Collapse"

Team β : Vertical Slicing by Sub-Problem

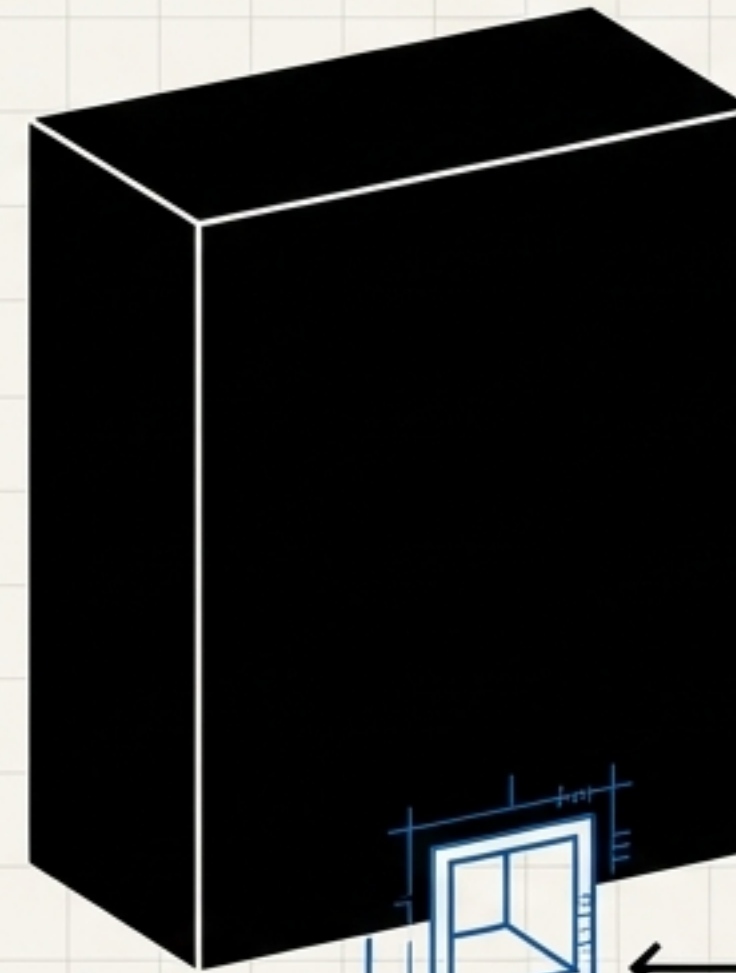


The Power of Information Hiding

Without Information Hiding

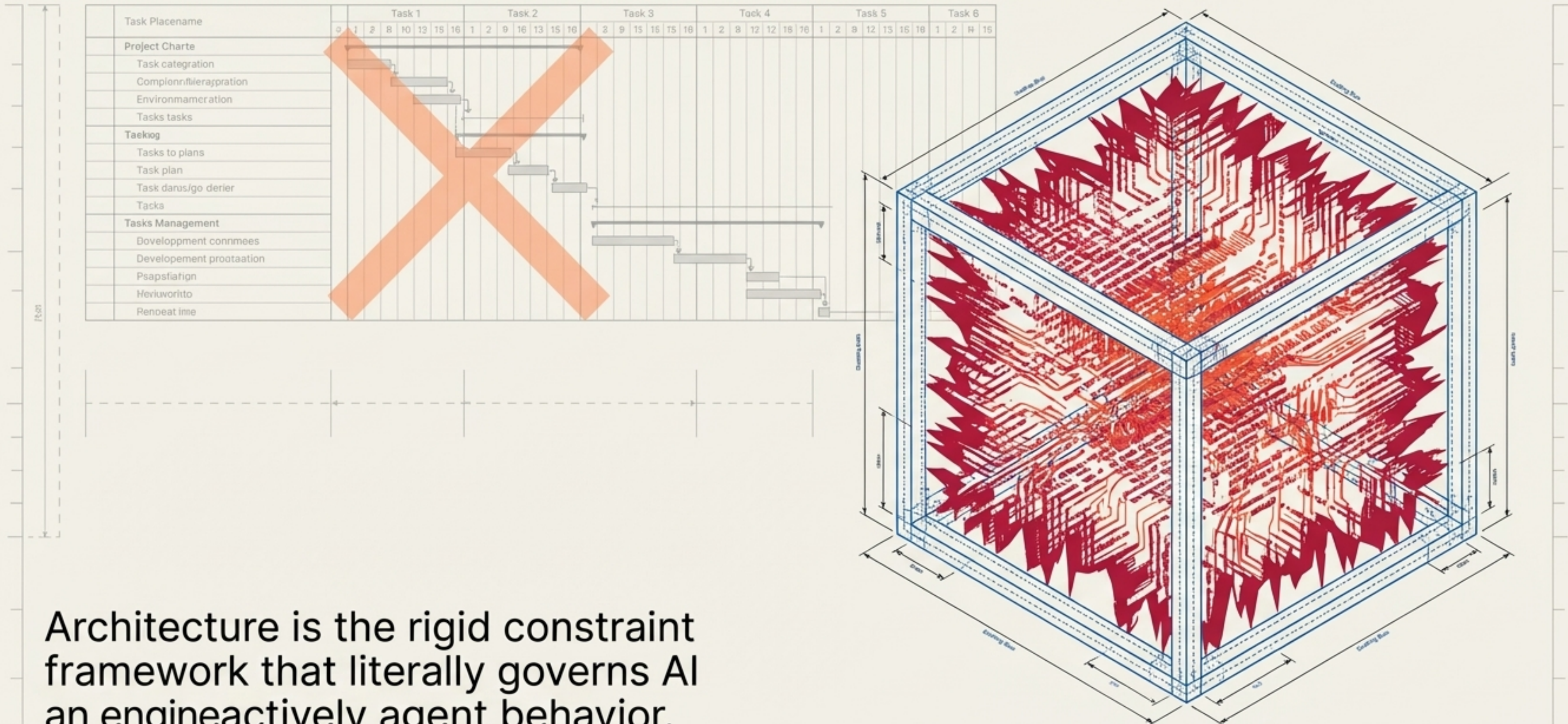


With Information Hiding



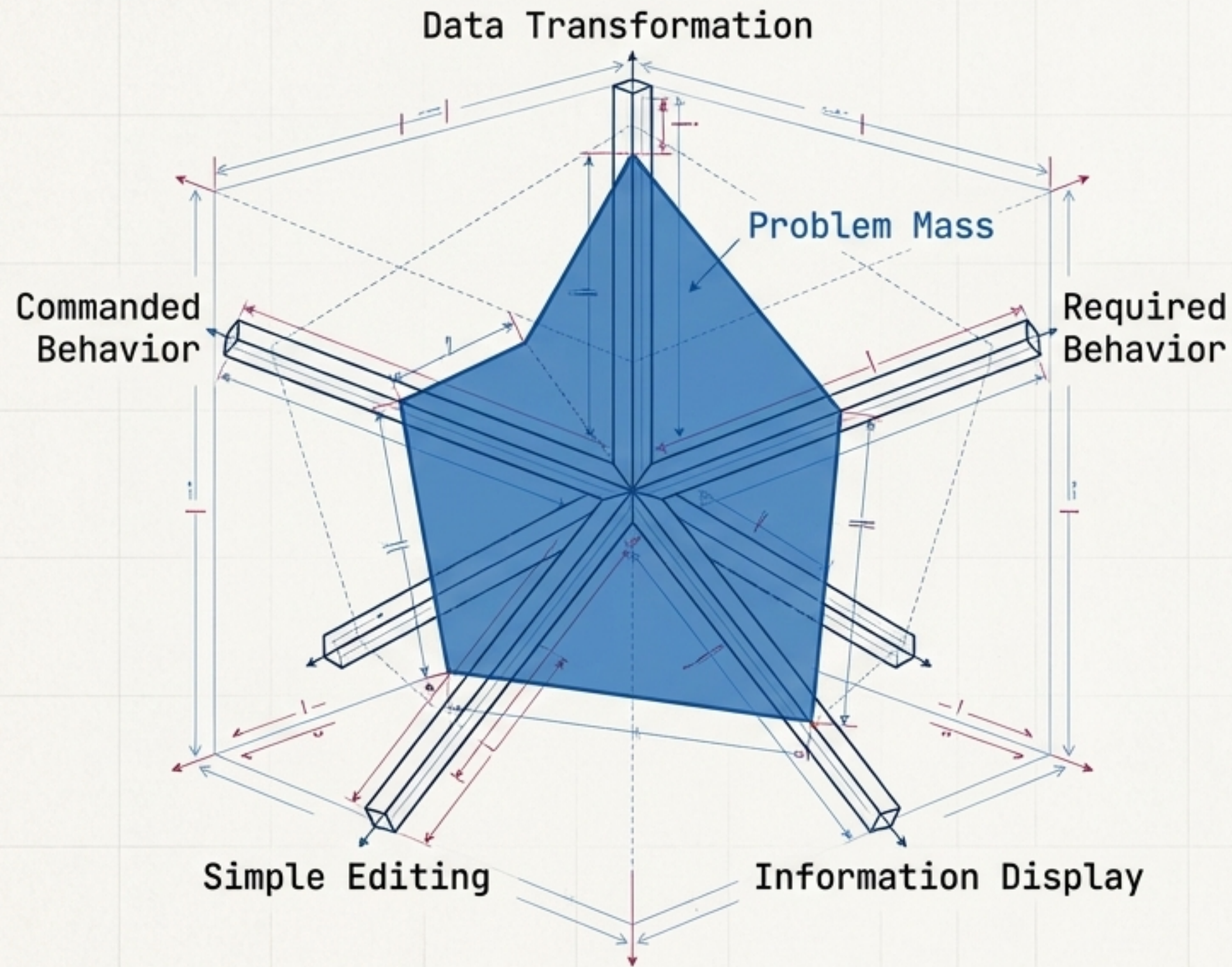
← **The API Contract:** What happens inside the stack is irrelevant. Teams and AI only need to respect shared interfaces.

Architecture is a Design Philosophy



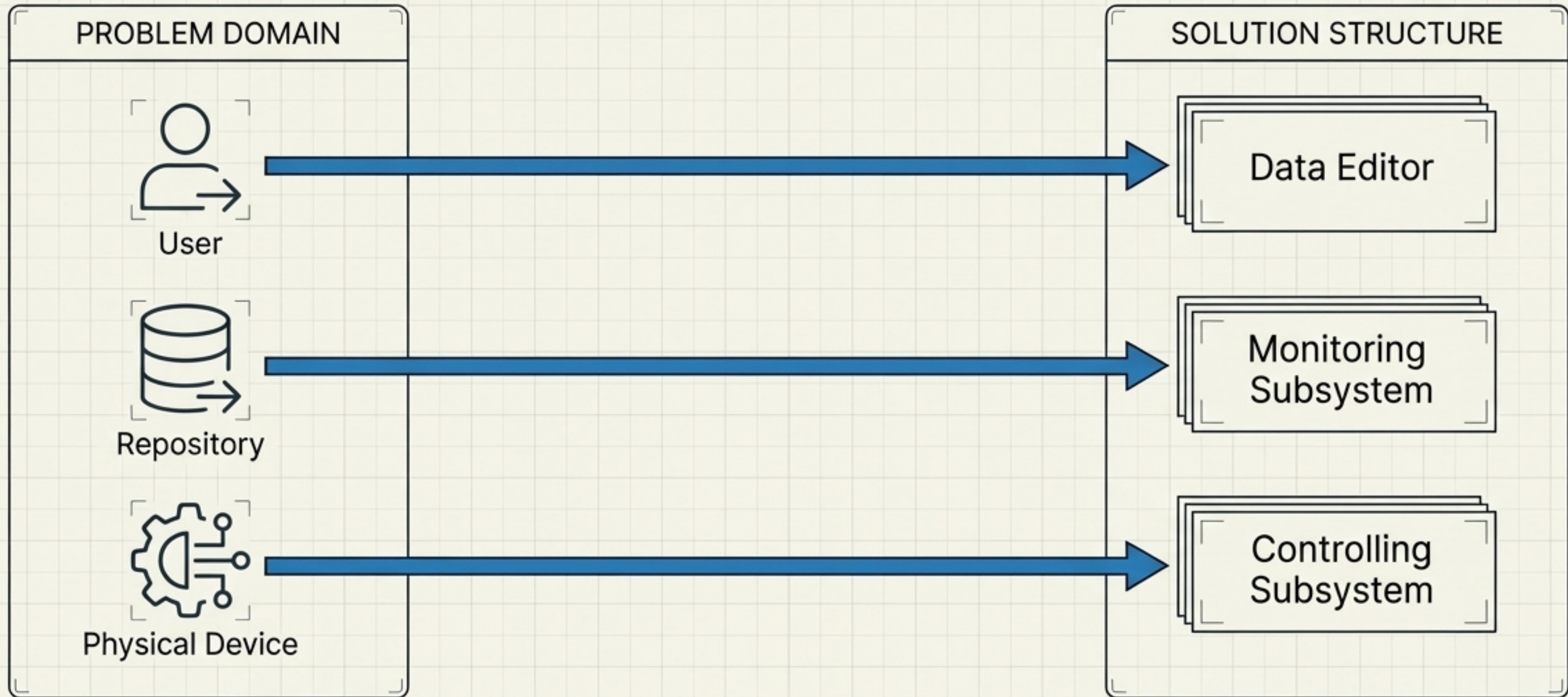
Architecture is the rigid constraint framework that literally governs AI an engineactively agent behavior.

The 5-Dimensional Problem Space



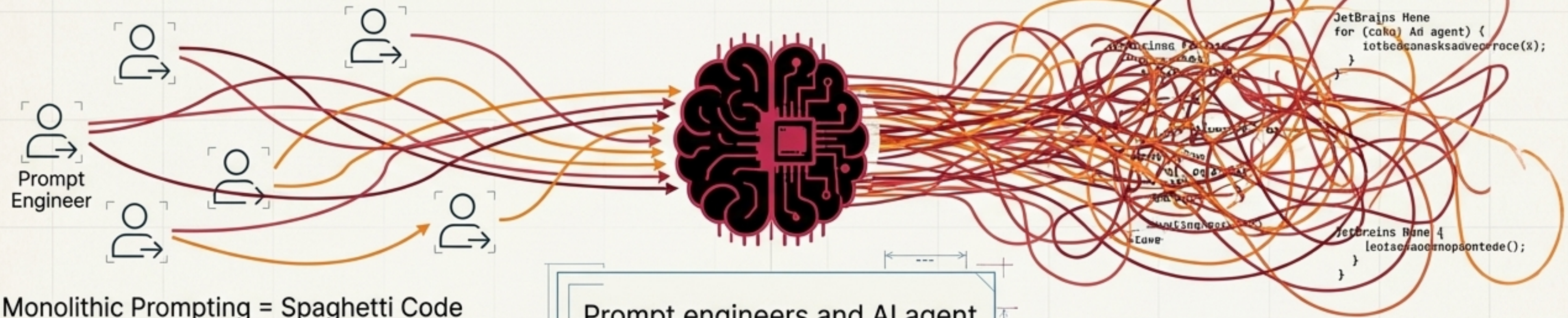
Every raw requirement can be mathematically mapped and decomposed along these five problem axes. This spatial structure directly dictates the AI prompt.

Aligning Solution to Problem Structure



The structure of the solution **MUST** be selected to perfectly fit the structure of the problem.

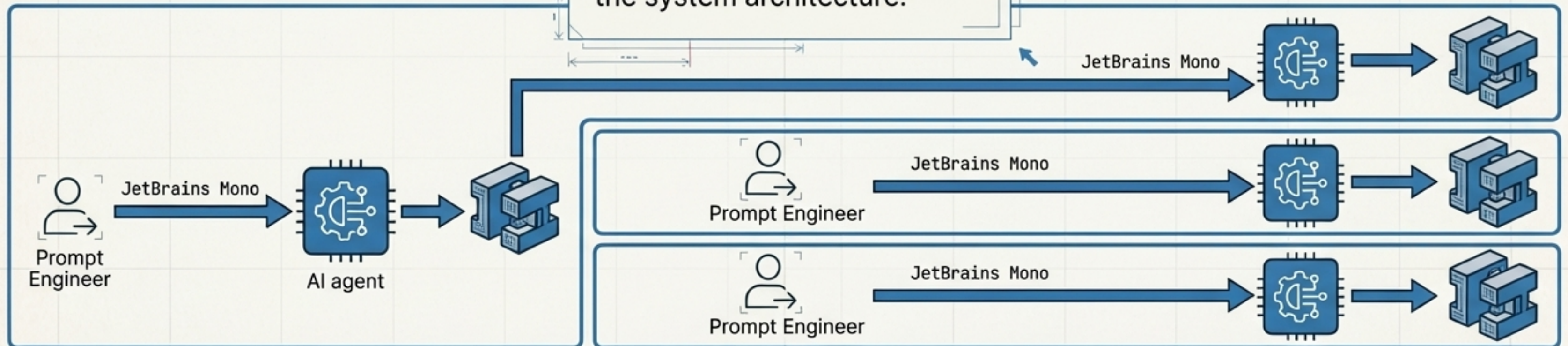
Conway's Law 2.0



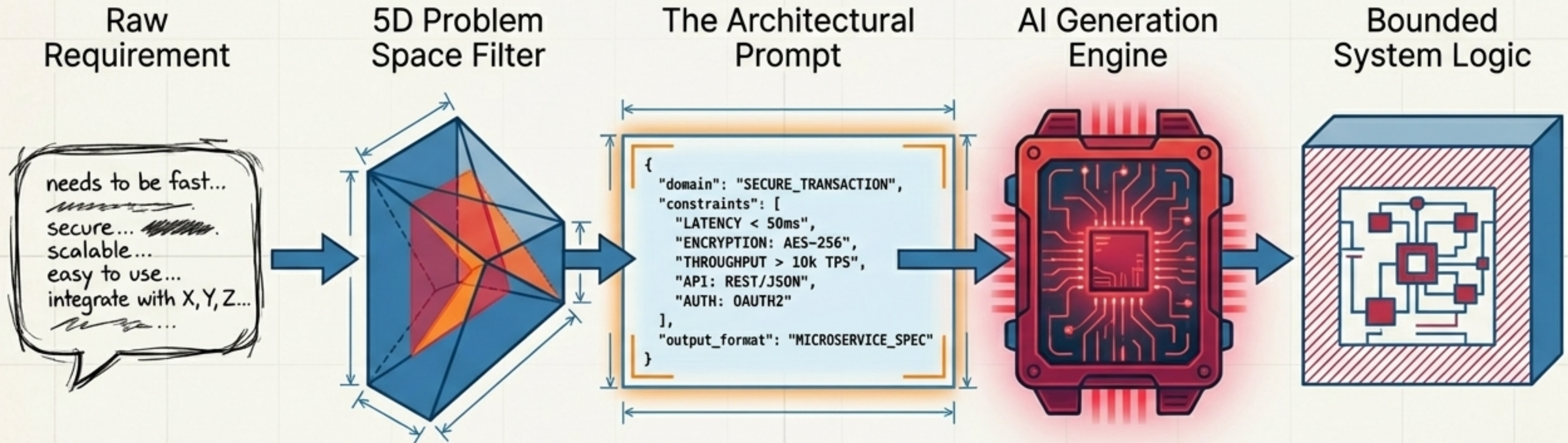
Monolithic Prompting = Spaghetti Code

Architectural Prompting = Clean Modularity

Prompt engineers and AI agent swarms will inevitably mirror the system architecture.

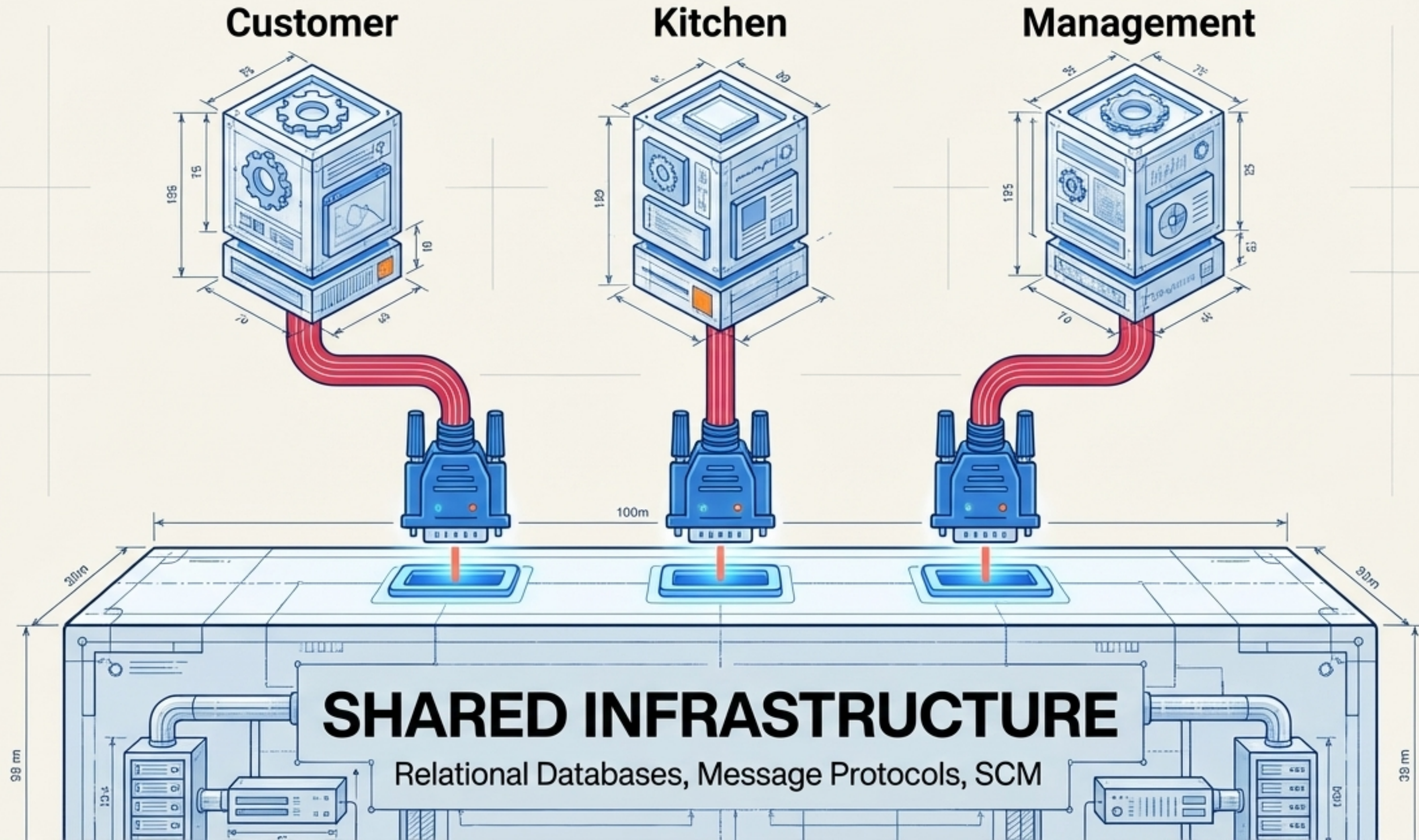


Architecture as a Prompt Engine



Raw requirements are mathematically filtered to generate strictly bounded architectural constraints. The AI generates logic solely within the black box.

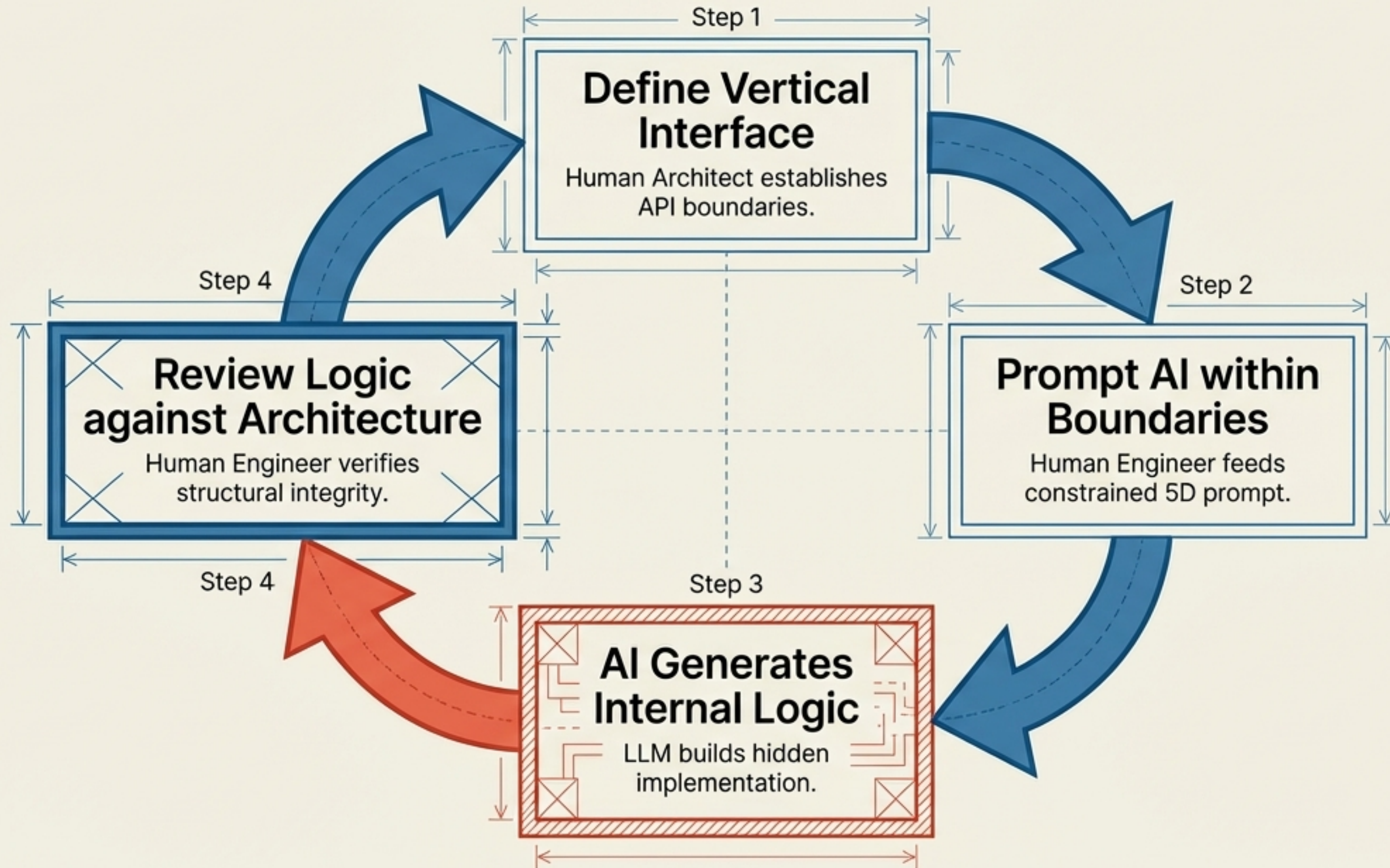
Establishing the Shared Infrastructure



The Work Division Matrix: AI Readiness

	Team α (Horizontal / Expertise)	Team β (Vertical / Problem)
Feedback Loop	Long (Depends on slowest link) JetBrains Mono, Inter	Short (Immediate per stack) JetBrains Mono, Inter
Failure Resilience	Project fails if one link fails	Stack succeeds independently
Communication	High (Endless documentation)	Low (API Contracts only)
AI-Readiness JetBrains Mono, Inter	Fails (Context window collapse) 	Succeeds (Bounded generation) 

The AI Integration Loop



The Paradigm Shift

Classic Software Engineering

Phase Focus:
Implementation

Core Skill:
Syntax Mastery

Primary Artifact:
Source Code

Human Role:
The Typist

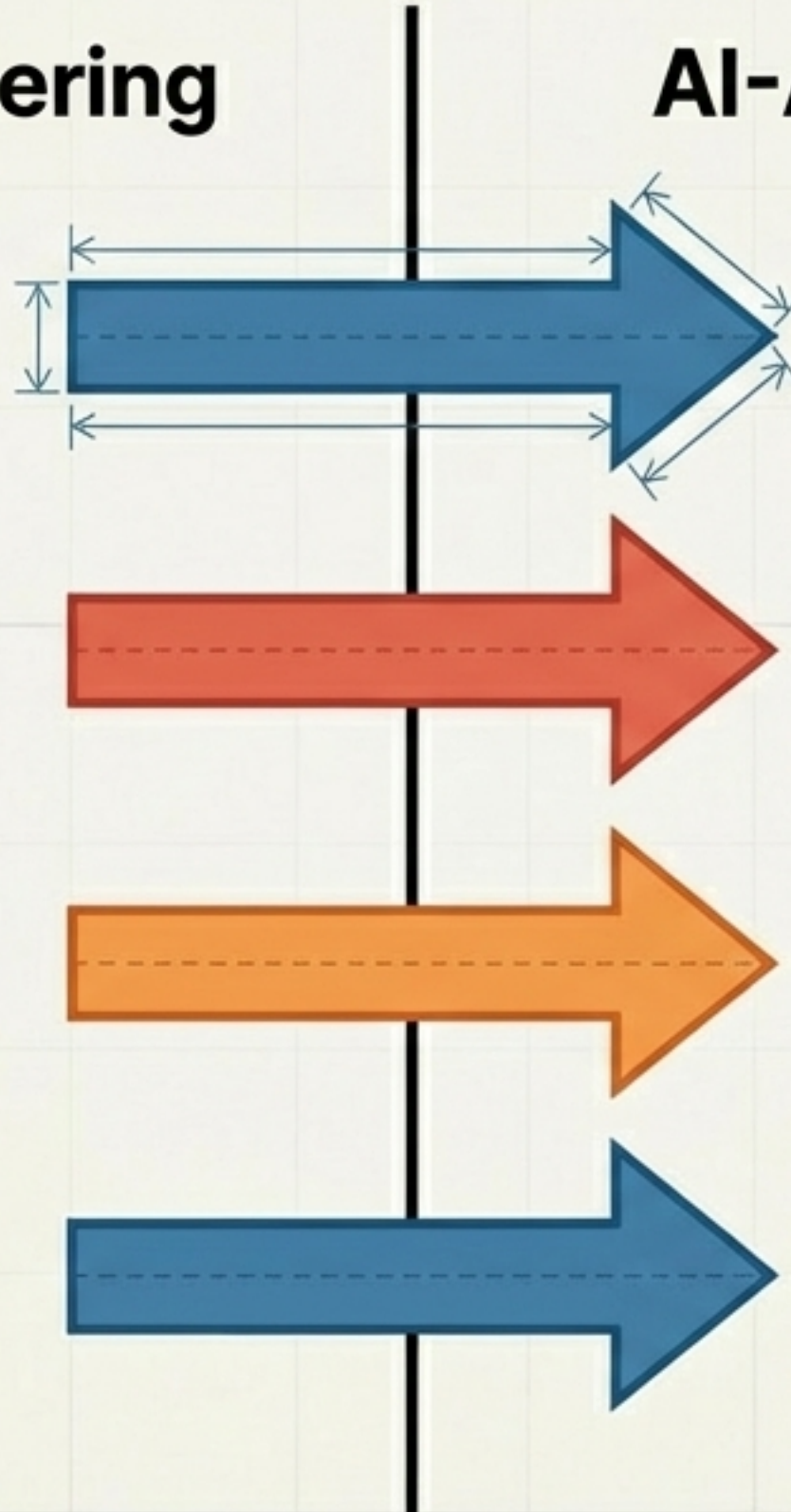
AI-Augmented Engineering

Phase Focus:
Architectural Prompting

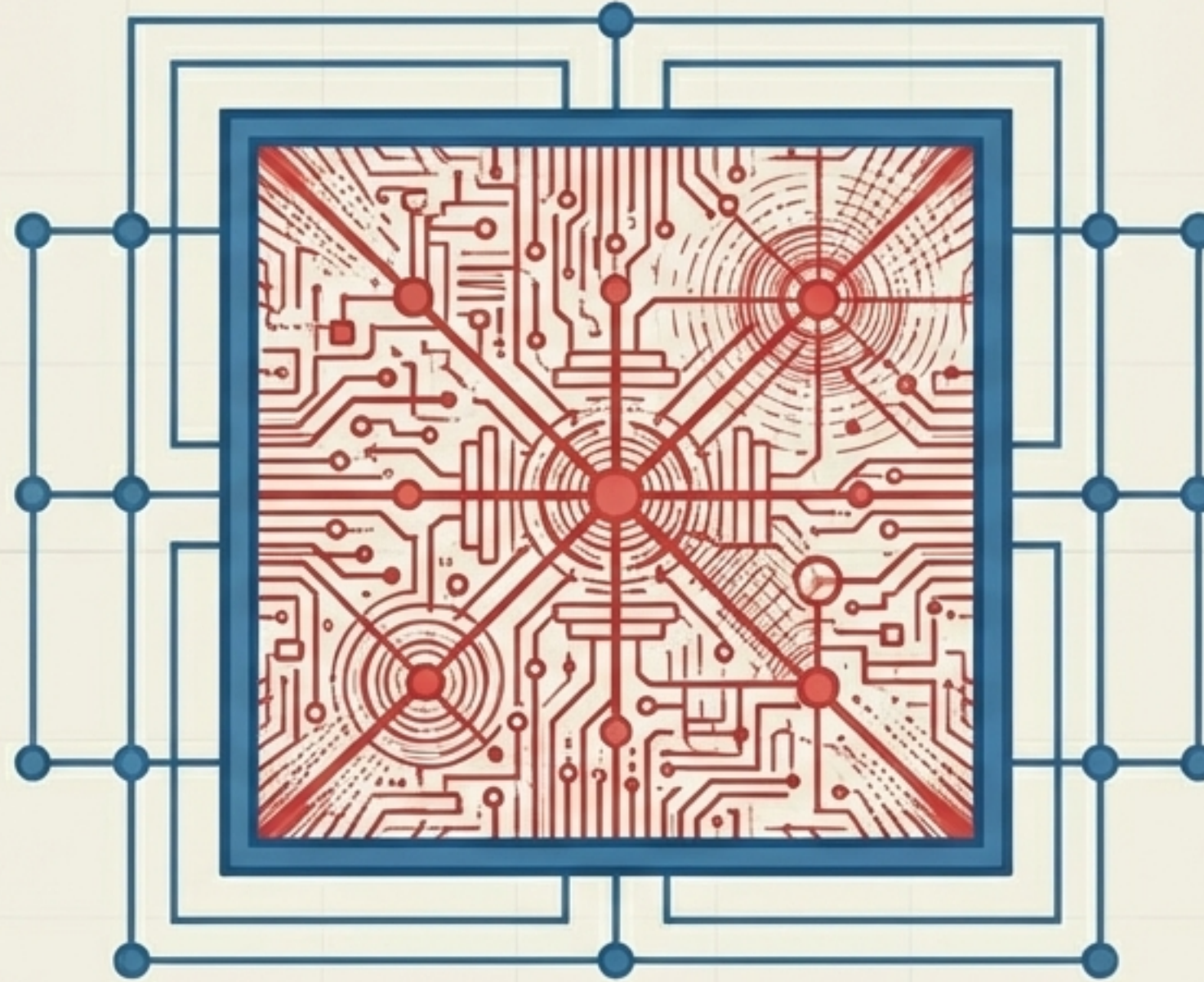
Core Skill:
Logic Review & Verification

Primary Artifact:
API Contracts & Boundaries
JetBrains Mono

Human Role:
The Orchestrator



The New Division of Labor



The Problem provides the purpose.
The **AI** provides the **labor**.
The Architect provides the boundaries.